Biologics Plant KPIs (simulated data)

Year Selector

2025

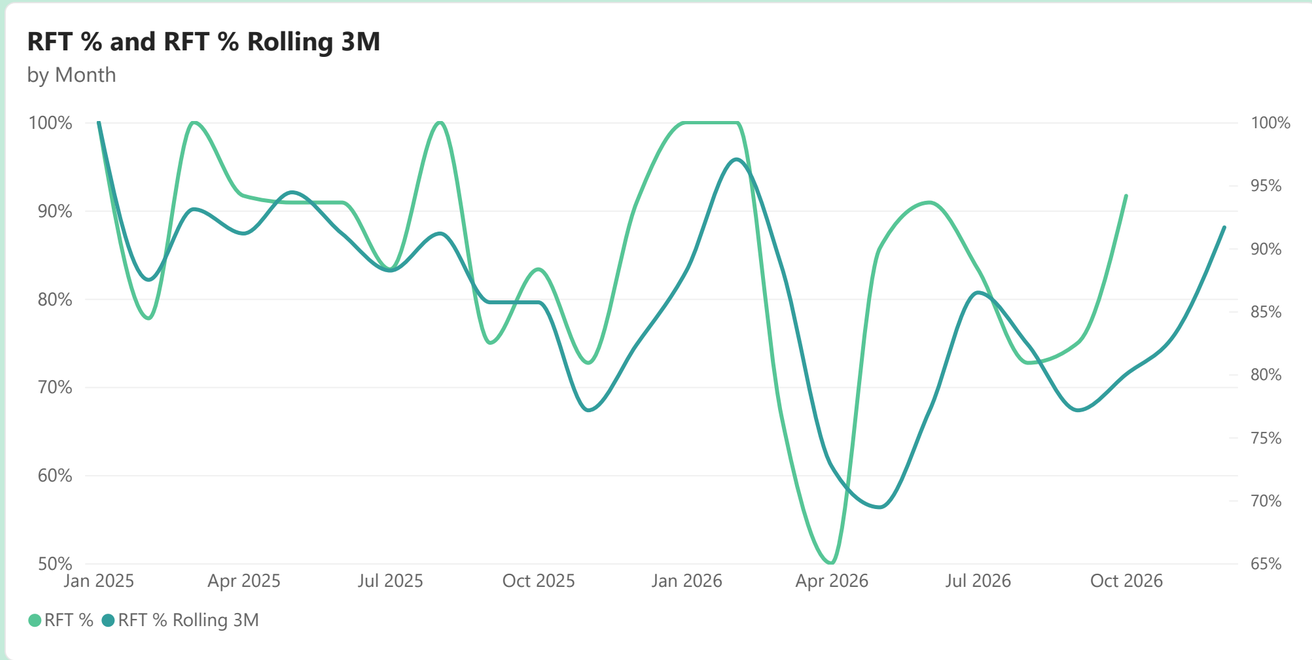
2026

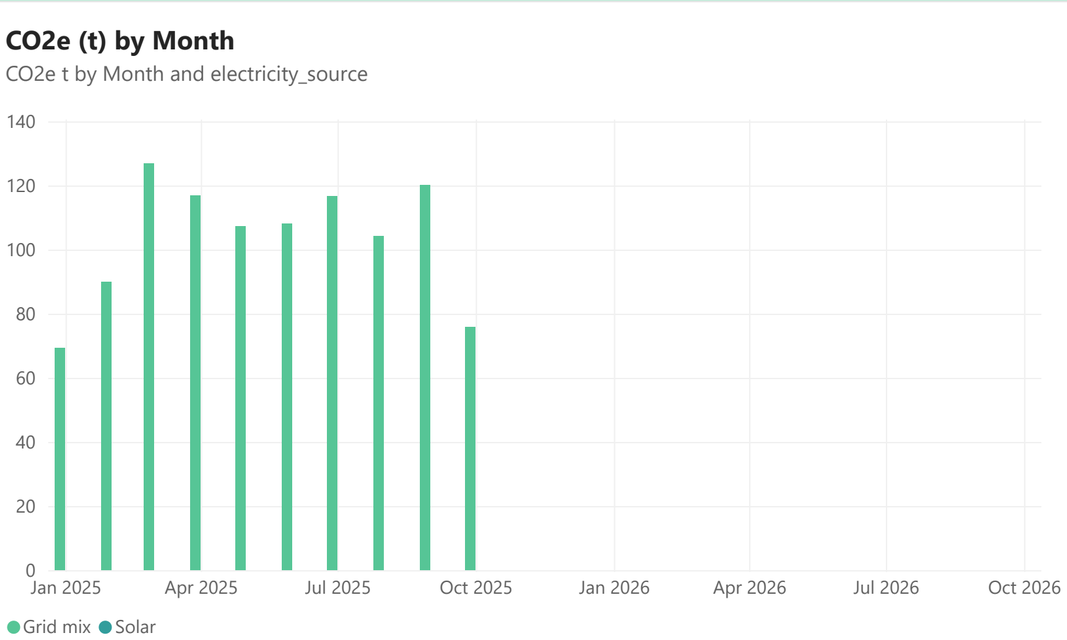
DISCLAIMER:
This is an exploration of simulated data calibrated to be comparable to a public FUJIFILM Hillerød flyer. This is not representative of FUJIFILM data.

By: Anthony Hobson

Key Findings:

- **A headline KPI can mask a quality problem**
  - Batch success is 98.3% YTD, but RFT has fallen from 87.9% to 82.1%, so RFT needs to be reported alongside batch success in a review setting.
- **A single asset was behind the RFT decline**
  - BR-07's RFT is 15% since Sep 2025, comparing to 87% for the other tanks ($p < 0.001$), and 197 h of agitator faults make it a target for maintenance.
- **A December QA backlog caused late deliveries**
  - OTD fell to 45% in Dec 2025 because QA release took about 6 days longer than usual, meaning QA capacity should be planned around the holidays.
- **Data validation prior to reporting**
  - R found and fixed duplicate rows, inconsistent product names, mixed date formats and missing titers, and every KPI in Power BI matches the R output exactly.





Batch Success % YTD 98.29%	RFT % 85.14%	OTD % 97.53%
Titer vs Target % 99.61%	Unplanned Downtime h 607.10	Solar Share % 58.06%

Bioreactor Deep-Dive

Year Selector

2025

2026

Product Selector

- [ ] Fc-412
- [ ] mAb-101
- [ ] mAb-205
- [ ] mAb-310
- [ ] mAb-518
- [ ] mAb-623

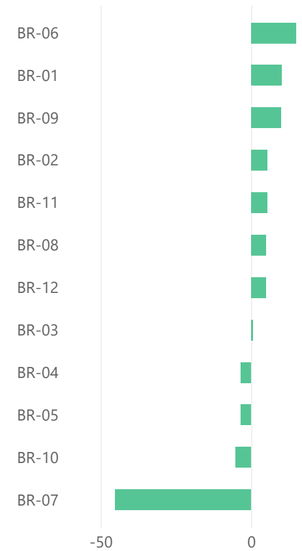
*The names listed are completely random and are not intended to reflect any actual products.

Worst Bioreactor (RFT)

BR-07

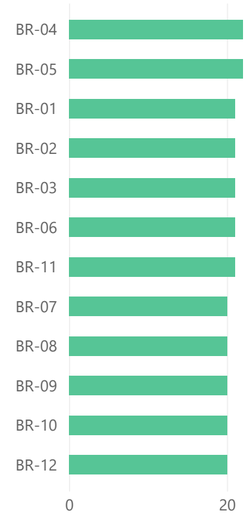
RFT Gap vs Site (pts)

by bioreactor



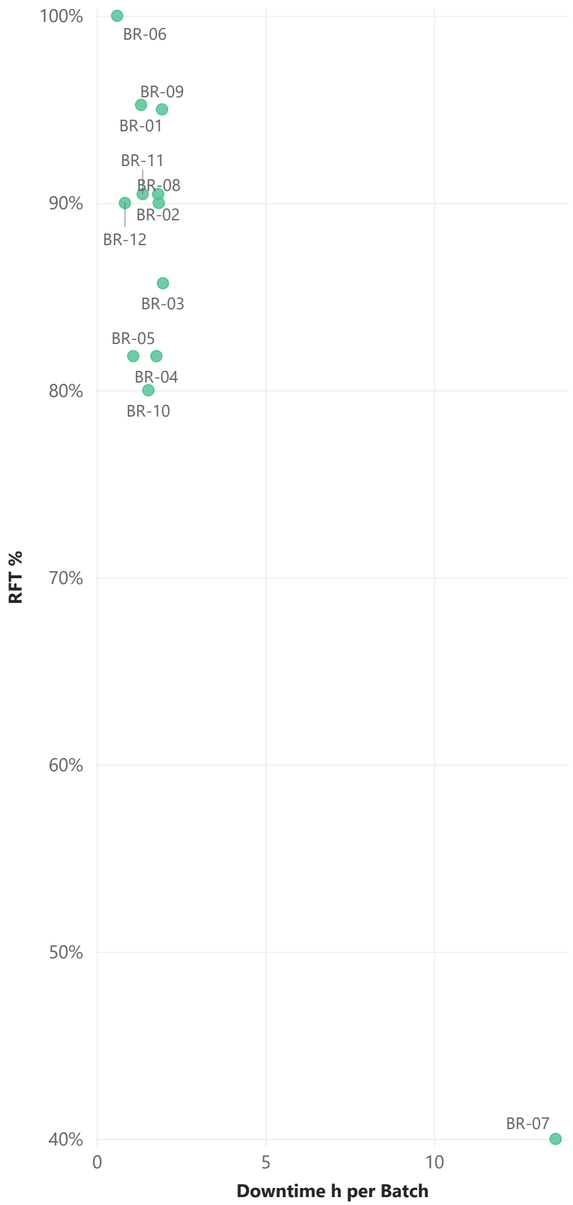
Count of run_length_band

by bioreactor



Downtime h per Batch and RFT %

by bioreactor



RFT Heat Map by Bioreactor

bioreactor	2025-Q1	2025-Q2	2025-Q3	2025-Q4	2026-Q1	2026-Q2	2026-Q3	2026-Q4
BR-01	100.00%	100.00%	66.67%	100.00%	100.00%	100.00%	100.00%	100.00%
BR-02	100.00%	100.00%	100.00%	66.67%	100.00%	66.67%	100.00%	100.00%
BR-03	100.00%	100.00%	50.00%	66.67%	100.00%	100.00%	66.67%	100.00%
BR-04	50.00%	66.67%	66.67%	100.00%	100.00%	66.67%	100.00%	100.00%
BR-05	100.00%	100.00%	100.00%	100.00%	100.00%	66.67%		100.00%
BR-06	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%
BR-07	100.00%	66.67%	66.67%		50.00%			100.00%
BR-08	100.00%	100.00%	100.00%	100.00%	66.67%	100.00%	50.00%	100.00%
BR-09	100.00%	100.00%	100.00%	66.67%	100.00%	100.00%	100.00%	100.00%
BR-10	50.00%	66.67%	100.00%	100.00%	100.00%	66.67%	100.00%	
BR-11	100.00%	100.00%	100.00%	100.00%	66.67%	66.67%	100.00%	100.00%
BR-12	100.00%	100.00%	66.67%	100.00%	66.67%	100.00%	100.00%	100.00%

Downtime (h) Matric by Bioreactor

bioreactor	Agitator fault	Control system	Sensor failure	Valve leak	**Total**
BR-01	12.00		6.00	9.70	**27.70**
BR-02	9.70	4.30	13.30	11.00	**38.30**
BR-03	5.20	4.20	12.20	19.70	**41.30**
BR-04	17.90		18.00	3.10	**39.00**
BR-05	13.80	2.30	7.80		**23.90**
BR-06			7.00	5.80	**12.80**
BR-07	197.20	24.20	16.20	34.60	**272.20**
BR-08	8.00	2.10	5.10	21.70	**36.90**
BR-09	10.50	6.40	14.90	7.00	**38.80**
BR-10	14.30	6.40	5.80	4.20	**30.70**
BR-11		5.80	4.30	18.60	**28.70**
BR-12	6.40		5.70	4.70	**16.80**
Total	**295.00**	**55.70**	**116.30**	**140.10**	**607.10**

Definitions and Reconciliation

Anthony Hobson
November, 2026

KPI Dictionary

KPI	Definition
Batch success %	Batches not rejected ÷ all batches
CO2e (t)	Grid kWh × 0.12 kg CO2e/kWh (assumed); solar = 0 (market-based scope 2)
On-time delivery %	Delivered on or before the promised date ÷ delivered batches (rejected batches excluded)
Right first time %	Released with 0 deviations ÷ all batches
Solar share %	Solar kWh ÷ total kWh
Titer vs target	Actual titer ÷ target titer, averaged (missing titers excluded)
Unplanned downtime (h)	Hours of unplanned equipment stops

Data Cleaning in R

issue	rows_affected	fix
Missing actual titer	2	Kept batch; excluded from yield KPI only
Product name with wrong case or extra spaces	3	Standardised to official name
Duplicate batch rows	4	Removed
Rows before vs after cleaning	4	253 rows in, 249 unique batches out
Start date typed as dd/mm/yyyy	4	Parsed both formats to one date type

Power BI (DAX)

Year	Batches	Batch Success %	RFT %	OTD %
2025	132	96.97%	87.88%	95.31%
2026	117	98.29%	82.05%	100.00%
Total	**249**	**97.59%**	**85.14%**	**97.53%**

R Findings

period	batches	batch_success_rate	rft_rate	on_time_delivery
2025	132	96.97%	87.88%	95.31%
2026	117	98.29%	82.05%	100.00%
All (Jan 2025 - Oct 2026)	249	97.59%	85.14%	97.53%

About this data:

- Simulated data, generated with Claude Opus 5.5
- Calibrated to public headline figures in a FUJIFILM Hillerød flyer
  - 12 × 20,000 L bioreactors
  - 98.5% batch success YTD
  - 100% solar electricity from October 2025
- None of the figures are FUJIFILM's

Assumptions:

- CO2e uses 0.12 kg/kWh for grid electricity and 0 for solar
- Run times and turnaround times are assumed based on research
- The product slicer doesn't filter downtime, as downtime isn't recorded per product.

Tools utilized:

- R (tidyverse) for data cleaning
- Power Query (M) for typing and shaping
- DAX for measures and KPI
- Power BI for creating this dashboard